A Semester Project Report on

Risk-Aware Portfolio Optimization System for Stock Trading Using Ensemble Machine Learning

Submitted in partial fulfilment for the
degree of Bachelor of Technology in
Information Technology

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Abstract

Stock market investment involves significant uncertainty, volatility, and financial risk, making accurate decision-making challenging for investors. Traditional stock prediction systems mainly focus on forecasting individual stock prices using statistical or machine learning models. However, these approaches often ignore portfolio-level interactions between assets and provide limited insight into investment risk.

This project proposes a **Risk-Aware Portfolio Optimization System for Stock Trading using Ensemble Machine Learning**. The system focuses on recommending optimal portfolio allocation across multiple assets while considering both risk and return. Historical stock market data of NIFTY 50 companies is collected using the Yahoo Finance API and processed to generate technical indicators and financial metrics.

The system is implemented using a Flask-based backend and React frontend connected through REST APIs. Users can register, complete KYC verification, select stocks, choose investment amount and risk preference, and receive optimized portfolio recommendations through an interactive dashboard.

The framework incorporates portfolio optimization using **Modern Portfolio Theory (MPT)** and applies risk metrics such as volatility, Sharpe ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), beta, drawdown, diversification score, and sector allocation analysis. The system also implements an ensemble machine learning pipeline consisting of **Random Forest, Ridge Regression, and HistGradientBoosting** models to generate BUY, SELL, or HOLD trading signals.

An interactive dashboard is developed to visualize portfolio allocation, benchmark comparison, portfolio growth, sector distribution, risk contribution, and trading signals. The developed prototype provides a transparent and risk-aware platform for portfolio management and stock trading.

Keywords: Stock Market Prediction, Portfolio Optimization, Ensemble Machine Learning, Risk Analysis, Explainable Artificial Intelligence (XAI), Modern Portfolio Theory, Financial Time-Series Analysis, Algorithmic Trading.

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Nomenclature

P_t	Price of a stock at time t
r_t	Return of a stock at time t
μ	Mean (expected) return of an asset
Σ	Covariance matrix of asset returns
w_i	Weight of asset i in the portfolio
R_p	Expected return of the portfolio
σ_p	Portfolio volatility (risk)
R_f	Risk-free rate of return
VaR	Value at Risk
CVaR	Conditional Value at Risk
RSI	Relative Strength Index
SMA	Simple Moving Average
MPT	Modern Portfolio Theory
SLSQP	Sequential Least Squares Programming optimization algorithm
ML	Machine Learning
XAI	Explainable Artificial Intelligence
API	Application Programming Interface

Chapter 1

Introduction

The stock market plays an important role in the global financial system by enabling individuals and institutions to invest in companies and generate returns on capital. However, stock market behavior is highly complex and influenced by factors such as economic conditions, company performance, market sentiment, and global events. Because of this complexity, predicting stock price movements and making effective investment decisions remains a challenging task for investors and financial analysts.

Traditionally, stock market analysis relied on statistical models and financial theories such as Modern Portfolio Theory (MPT), Capital Asset Pricing Model (CAPM), and Efficient Market Hypothesis (EMH) to understand market behavior and guide investment decisions. These models introduced important concepts such as diversification and risk management but often rely on simplified assumptions and may not fully capture the dynamic nature of financial markets.

With the advancement of computing power and the availability of large financial datasets, machine learning techniques have increasingly been applied to stock market prediction and analysis. Machine learning models such as Random Forests, Ridge Regression, and Gradient Boosting are capable of identifying complex patterns in historical market data and generating predictive trading signals.

Despite these advancements, many existing stock prediction systems mainly focus on forecasting the future price of individual stocks. In practical investment scenarios, however, investors are more interested in managing risk and optimizing the allocation of capital across multiple assets rather than predicting the price of a single stock. Portfolio-level analysis is therefore essential for balancing risk and return in investment decisions.

Another limitation of many machine learning-based financial systems is the lack of transparency in their predictions. Advanced predictive models often operate as “black-box” systems, making it difficult for investors to understand the reasoning behind generated trading signals. In financial applications, where decisions involve significant monetary risk, explainability and transparency are important for building user trust.

To address these challenges, this project proposes a **Risk-Aware Portfolio Optimization System for Stock Trading using Ensemble Machine Learning**. The proposed system integrates ensemble machine learning models such as Random Forest, Ridge Regression, and HistGradientBoosting to generate BUY, SELL, and HOLD trading signals using historical market data and technical indicators. In addition, portfolio optimization techniques based on Modern Portfolio Theory are used to determine optimal asset allocation while minimizing portfolio risk. The system also incorporates risk analysis metrics such as volatility, Sharpe ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), beta, and diversification analysis. An interactive dashboard is developed to provide portfolio visualization, benchmark comparison, trading insights, and risk-aware investment decision support.

1.1 Objectives of the Study

- To develop a machine learning-based system for stock market analysis and portfolio optimization using historical financial data.
- To generate trading signals such as BUY, SELL, and HOLD to assist investors in investment decision-making.
- To optimize portfolio allocation using risk-aware strategies based on Modern Portfolio Theory (MPT).
- To analyze portfolio risk using financial metrics such as volatility, Sharpe ratio, Value at Risk (VaR), and diversification analysis.

- To implement ensemble machine learning models for improving prediction accuracy and trading signal generation.
- To provide portfolio visualization and performance analysis through an interactive dashboard.
- To provide explainable insights into model predictions using explainable artificial intelligence techniques.

1.2 Organization of the report

The report is organized into several chapters to present the study in a structured manner.

- Chapter 1 introduces the background, problem statement, objectives, and scope of the study.
- Chapter 2 presents the literature review, including the evolution of stock trading systems and analysis of existing platforms.
- Chapter 3 describes the proposed system architecture and methodology used in the project.
- Chapter 4 discusses the technology stack and implementation approach.
- Finally, Chapter 5 presents the expected results, future scope, and conclusion of the study.

Chapter 2

Review of Literature

This chapter discusses the existing research and solutions related to stock market analysis, trading systems, and portfolio optimization techniques. Several financial models, machine learning approaches, and trading platforms have been developed to assist investors in making informed decisions. However, many of these systems have certain limitations such as lack of risk awareness, limited portfolio optimization capabilities, and lack of explainability in machine learning models. This section reviews the evolution of stock trading prediction systems, existing financial platforms, and the models commonly used in stock market analysis.

2.1 Evolution of Stock Trading and Prediction Systems

Stock market prediction has been widely studied in financial research for several decades. Early research mainly relied on statistical and financial theories to explain market behavior and support investment decisions. One of the earliest contributions was Modern Portfolio Theory proposed by Markowitz, which introduced the concept of diversification to reduce investment risk by allocating capital across multiple assets (1).

Later, the Capital Asset Pricing Model (CAPM) developed by Sharpe explained the relationship between risk and expected return using the beta coefficient, which measures the sensitivity of a stock relative to the overall market (2). Similarly, the Efficient Market Hypothesis proposed by Fama suggested that stock prices reflect all available information, making consistent prediction of market movements difficult (3).

During the late twentieth century, researchers began using statistical time-series models such as ARIMA and GARCH to forecast stock price movements and volatility (4). However, these models assume linear relationships and may not capture the complex and nonlinear behavior of financial markets.

With the advancement of computing technology, machine learning techniques have been increasingly applied to stock market prediction. Artificial Neural Networks have been used to model nonlinear relationships between financial indicators and stock prices (5). Support Vector Machines have also been widely used for financial time-series classification and trend prediction (6). Later, ensemble models such as Random Forest were introduced to improve prediction stability by combining multiple decision trees (7). More recently, deep learning models such as Long Short-Term Memory (LSTM) networks have shown promising results in analyzing sequential financial data and capturing long-term market dependencies (8).

Although these approaches have improved prediction accuracy, most systems still focus on predicting individual stock prices rather than optimizing portfolio-level investment decisions.

2.2 Study of Existing Stock Trading Platforms

Modern financial markets rely on various digital platforms that support trading analysis, automated trading strategies, and portfolio management. One widely used platform is QuantConnect, which provides a cloud-based environment for algorithmic trading and quantitative research using programming languages such as Python and C#. While the platform offers extensive financial datasets and backtesting capabilities, it requires advanced programming knowledge and lacks explainability in machine learning-based strategies.

TradingView is another popular platform used for financial charting and technical analysis. It provides real-time stock charts and technical indicators for market analysis. However, the platform mainly focuses on technical indicators and does not integrate machine learning-based prediction models or portfolio optimization tools.

MetaTrader is widely used for automated trading through predefined strategies known as Expert Advisors. Although it supports market analysis and automated trading execution, it primarily relies on rule-based trading strategies rather than advanced machine learning approaches.

Other platforms such as Alpaca provide trading APIs for building automated trading systems, while platforms like Portfolio Visualizer focus on portfolio performance analysis and asset allocation strategies. Robo-advisor platforms such as Wealthfront and Betterment automatically manage investment portfolios based on predefined asset allocation strategies.

Despite the availability of these platforms, most existing systems focus on individual aspects such as chart analysis, automated trading, or portfolio evaluation rather than integrating machine learning prediction, risk analysis, and portfolio optimization into a unified system (9). Furthermore, many platforms lack explainability and transparency in their trading decisions.

2.3 Models Used in Stock Trading and Portfolio Optimization

Various models from statistics, finance, and machine learning have been used to analyze financial markets and support trading decisions. Volatility models such as GARCH are commonly used to estimate market risk and analyze changes in financial volatility over time (10). Technical indicators such as Moving Averages, MACD, and Relative Strength Index (RSI) are widely used to detect market trends and momentum patterns.

Machine learning models such as Random Forest and Logistic Regression are commonly used for classification problems in financial prediction, including predicting whether stock prices will increase or decrease. Random Forest improves prediction accuracy by combining multiple decision trees and is particularly effective when dealing with noisy financial data (7).

Deep learning models such as Long Short-Term Memory networks have also been applied to financial time-series prediction due to their ability to learn sequential dependencies in historical stock price data (8). In addition to prediction models, portfolio optimization techniques play an important role in investment decision making.

Modern Portfolio Theory introduced by Markowitz provides a mathematical framework for balancing risk and return by diversifying investments across multiple assets (1). Risk-adjusted performance metrics such as the Sharpe Ratio are widely used to evaluate the efficiency of investment portfolios (2). Other risk measures such as maximum drawdown and Value at Risk (VaR) are used to estimate potential losses in investment portfolios.

Although these models provide valuable insights into financial market behavior, many existing systems still focus on price prediction rather than portfolio-level risk management. Therefore, integrating machine learning prediction models with portfolio optimization and explainable artificial intelligence can provide a more reliable decision-support system for stock trading.

From Tables 2.1 and 2.2, it can be observed that most existing platforms focus on specific aspects such as algorithmic trading, portfolio analysis, or social trading. However, these platforms rarely integrate machine learning prediction, risk-aware portfolio optimization, and explainable artificial intelligence in a single framework. Many platforms also operate as black-box systems and lack transparency in their decision-making processes. These limitations highlight the need for a unified system that combines machine learning models, portfolio optimization techniques, and explainable AI to support reliable and transparent stock trading decisions.

Table 2.1: Comparison of Algorithmic and AI-based Trading Platforms

Platform	**Main Features**	**Strengths**	**Limitations**
QuantConnect	Cloud-based algorithmic trading platform supporting Python and C# with historical datasets and backtesting tools.	Provides extensive financial datasets and supports quantitative trading research.	Requires strong programming skills and lacks explainability in machine learning trading decisions.
TradingView	Web-based financial charting platform providing real-time stock charts and technical indicators.	Powerful visualization and technical analysis tools widely used by traders.	Focuses mainly on chart-based analysis and does not support machine learning prediction or portfolio optimization.
FinRL	Open-source reinforcement learning framework designed for financial trading research.	Enables development of AI-based trading agents using deep reinforcement learning.	Requires advanced machine learning knowledge and lacks explainable insights for predictions.
MetaTrader (MT4/MT5)	Automated trading platform supporting Expert Advisors for rule-based strategies.	Supports automated trading execution and strategy backtesting.	Primarily uses rule-based trading strategies and lacks machine learning integration.
Alpaca	Trading API platform for building algorithmic trading systems using programming languages like Python.	Provides real-time market data and infrastructure for automated trading.	Does not include built-in machine learning prediction models or portfolio optimization features.
Zipline	Open-source backtesting library used for quantitative finance research.	Useful for testing trading strategies using historical market data.	Limited integration with modern machine learning pipelines and real-time trading systems.

Table 2.2: Comparison of Portfolio Management and Investment Platforms

Platform	**Main Features**	**Strengths**	**Limitations**
Portfolio Visualizer	Financial research platform used for portfolio performance analysis and asset allocation strategies.	Provides tools for analyzing risk-adjusted returns and portfolio optimization.	Relies mainly on traditional financial models and lacks machine learning prediction capabilities.
eToro	Social trading platform allowing investors to copy trading strategies of experienced traders.	Easy to use and accessible for beginner investors.	Trading decisions rely on user behavior rather than data-driven machine learning analysis.
Wealthfront	Robo-advisor platform that automatically manages investment portfolios.	Provides automated portfolio rebalancing and tax-efficient investment strategies.	Uses predefined portfolio models and lacks transparency in decision-making algorithms.
Betterment	Automated investment platform that allocates assets based on investor risk profiles.	Simplifies investment management and provides diversified portfolios.	Limited customization and lacks machine learning-based prediction models.
Proposed System	Risk-aware portfolio optimization system using ensemble machine learning models and explainable AI techniques.	Integrates prediction models, risk analysis, portfolio optimization, and explainable AI within one system.	Currently designed as a research prototype and requires further development for real-time trading deployment.

Chapter 3

Methodology

This chapter describes the methodology used for developing the Risk-Aware Portfolio Optimization System. The system follows a structured workflow for stock portfolio optimization, risk analysis, trading signal generation, and ensemble machine learning prediction.

3.1 User Authentication and KYC Verification

The system first handles user authentication and identity verification before allowing access to portfolio analysis features.

3.1.1 Signup Process

The signup module allows users to create accounts using their name, email address, and password.

The following steps are performed:

- User enters registration details
- Password is securely hashed before storage
- User information is stored in PostgreSQL database
- JWT token is generated after successful signup

3.1.2 KYC Verification

After signup, users complete KYC verification by submitting:

- PAN Card
- Aadhaar Card
- Selfie Upload

The backend validates the JWT token and updates the KYC status in the database.

Only KYC-approved users are allowed to access the dashboard.

3.1.3 Login Process

The login module performs:

- Credential verification
- Password validation
- KYC approval checking
- JWT token generation

Users with incomplete KYC verification are restricted from accessing portfolio optimization features.

3.2 Data Collection

Historical stock market data is collected using the Yahoo Finance API through the `yfinance` Python library.

The system:

- Downloads approximately two years of historical stock price data
- Supports NIFTY 50 stocks

- Removes invalid or unavailable ticker symbols automatically

The Adjusted Closing Price is primarily used for calculations and portfolio analysis.

3.3 Data Preprocessing

Before optimization, preprocessing is applied to the collected stock market data.

The preprocessing stage performs the following tasks:

- Aligns stock data on common trading dates
- Removes missing values
- Filters incomplete stock records
- Calculates daily percentage returns

Daily return is calculated using:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{3.1}$$

where:

- P_t represents the current stock price
- P_{t-1} represents the previous stock price

The generated return series is used for risk analysis and optimization.

3.4 Risk Analysis

The system calculates several financial risk metrics to evaluate portfolio stability and performance.

3.4.1 Mean Return

The average return of each stock is calculated as:

$$\mu_i = \frac{1}{N} \sum_{t=1}^{N} r_{i,t} \tag{3.2}$$

where $r_{i,t}$ represents the return of stock i at time t.

3.4.2 Covariance Matrix

The covariance matrix measures how stock returns move relative to each other.

$$\Sigma_{ij} = \frac{1}{N-1} \sum_{t=1}^{N} (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j) \tag{3.3}$$

3.4.3 Portfolio Volatility

Portfolio volatility is calculated using:

$$\sigma_p = \sqrt{w^T \Sigma w} \tag{3.4}$$

where:

- w represents portfolio weights
- Σ represents the covariance matrix

3.4.4 Value at Risk (VaR)

Value at Risk estimates the maximum expected loss at a given confidence level.

$$VaR_{95} = -Q_{5\%}(R) \tag{3.5}$$

3.4.5 Conditional Value at Risk (CVaR)

Conditional Value at Risk estimates the average loss during worst-case scenarios.

$$CVaR_{95} = -E[R \mid R \leq Q_{5\%}(R)] \tag{3.6}$$

The system also calculates:

- Sharpe Ratio
- Beta
- Maximum Drawdown
- Risk Contribution

3.5 Portfolio Optimization

Portfolio optimization is implemented using Modern Portfolio Theory (MPT).

The optimization module receives:

- Selected stocks
- Investment amount
- User risk preference

The system calculates expected returns and covariance relationships between assets and generates optimal portfolio weights.

3.5.1 Portfolio Return

Expected portfolio return is calculated as:

$$R_p = \sum_{i=1}^{n} w_i \mu_i \tag{3.7}$$

where:

- w_i represents weight of asset i
- μ_i represents expected return of asset i

3.5.2 Optimization Strategies

The system supports different optimization modes based on user risk level:

- Low Risk – Minimize volatility
- Medium Risk – Maximize Sharpe Ratio
- High Risk – Maximize expected return

3.5.3 Optimization Constraints

The optimization problem is solved under the following constraints:

$$\sum_{i=1}^{n} w_i = 1 \tag{3.8}$$

$$0 \leq w_i \leq 1 \tag{3.9}$$

The Sequential Least Squares Programming (SLSQP) algorithm from SciPy is used to solve the optimization problem.

3.6 Trading Signal Generation

The system generates BUY, HOLD, and SELL signals using technical indicators.

3.6.1 Technical Indicators Used

- Relative Strength Index (RSI)
- 20-Day Simple Moving Average (SMA20)
- 50-Day Simple Moving Average (SMA50)
- Momentum Indicator

3.6.2 Simple Moving Average

The Simple Moving Average is calculated as:

$$SMA_n = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{3.10}$$

3.6.3 Relative Strength Index

The RSI is calculated using:

$$RS = \frac{\text{Average Gain}}{\text{Average Loss}} \tag{3.11}$$

$$RSI = 100 - \frac{100}{1 + RS} \tag{3.12}$$

3.6.4 Trading Rules

The following rules are used for signal generation:

- RSI $<$ 35 indicates BUY signal
- RSI $>$ 65 indicates SELL signal
- SMA20 $>$ SMA50 indicates bullish trend
- Positive momentum indicates upward trend

Signals from multiple indicators are combined to generate the final stock recommendation.

3.7 Portfolio-Level Signal Generation

Portfolio-level recommendations are generated using aggregate stock-level signals.

- If majority of stocks generate BUY signals, portfolio signal is BUY
- If majority of stocks generate SELL signals, portfolio signal is SELL
- Otherwise, portfolio signal is HOLD

3.8 Ensemble Machine Learning Module

Phase 2 introduces an ensemble machine learning pipeline for prediction and explainability.

3.8.1 Machine Learning Models Used

- Ridge Regression
- Random Forest
- HistGradientBoosting

3.8.2 Ensemble Workflow

The workflow includes:

- Feature extraction from stock data
- Model prediction using all three models
- Averaging predictions using ensemble learning
- Returning prediction explanations and contribution scores

The ensemble model improves prediction stability and reduces dependence on a single model.

3.9 Dashboard and Visualization

The frontend dashboard is developed using React and Recharts.

The dashboard provides:

- Portfolio allocation charts
- Sector allocation visualization
- Portfolio versus benchmark comparison

- Risk contribution charts
- Trading signal explanations
- Ensemble prediction insights

Users can interactively analyze portfolio performance and portfolio risk.

3.10 Benchmark Comparison

The optimized portfolio is compared against the NIFTY benchmark index.

Portfolio cumulative return is calculated using:

$$V_t = V_0 \prod_{i=1}^{t} (1 + r_i) \tag{3.13}$$

where:

- V_0 represents initial investment
- r_i represents daily return

This comparison helps evaluate portfolio performance against the market benchmark.

Chapter 4

Realisation and Implementation of the Proposed System

This chapter describes the practical implementation of the Risk-Aware Portfolio Optimization System. The system is implemented as a full-stack web application consisting of a Flask-based backend, React-based frontend, and PostgreSQL database support for authentication and user management.

The implementation integrates financial data retrieval, preprocessing, portfolio optimization, risk analysis, trading signal generation, ensemble machine learning prediction, and interactive dashboard visualization into a unified platform.

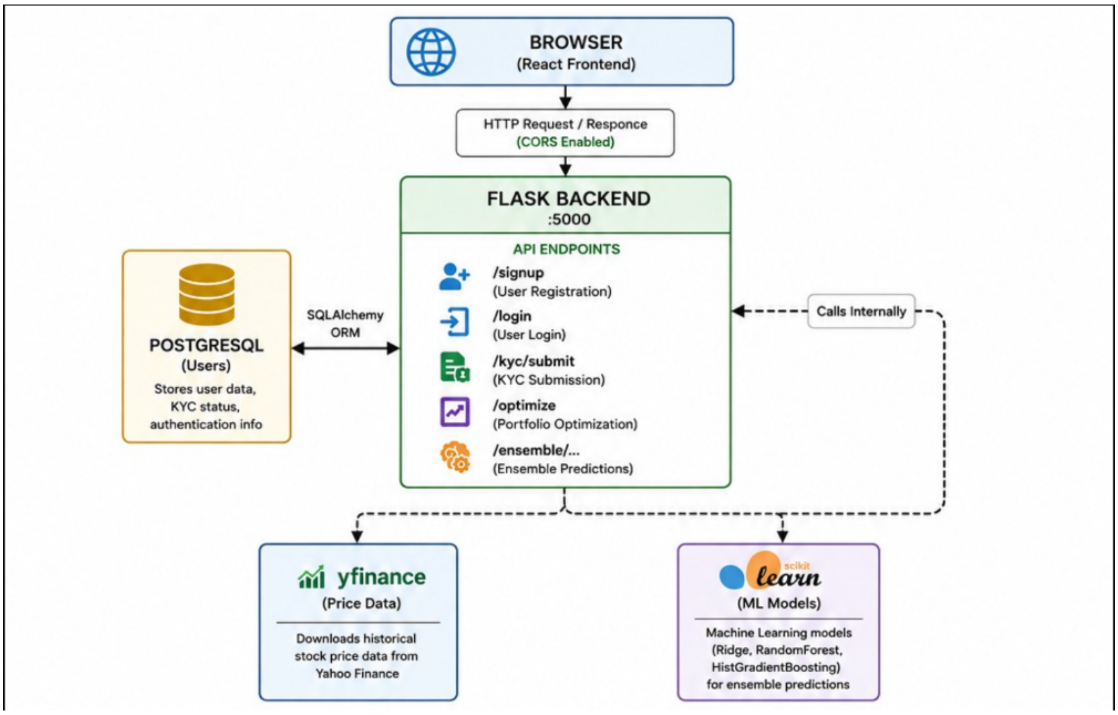

Figure 4.1: Overall System Architecture

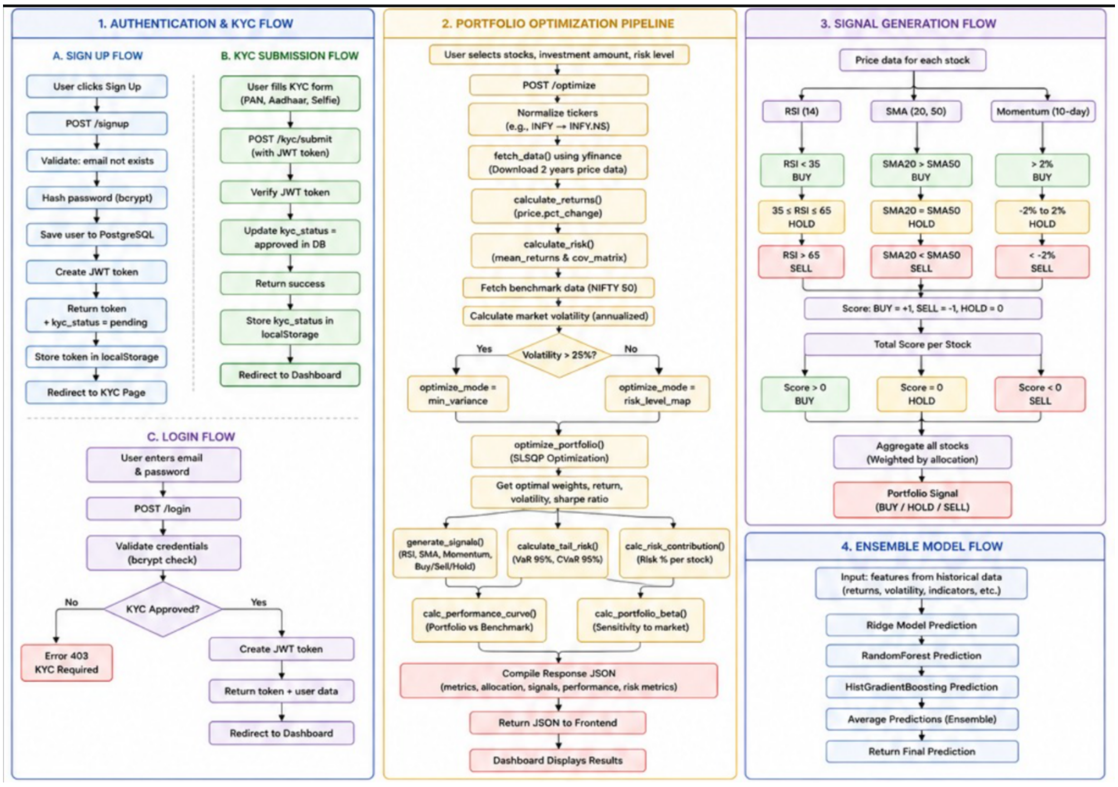

Figure 4.2: Detailed System Workflow

4.1 System Architecture

The system follows a client-server architecture consisting of multiple integrated modules for authentication, market data retrieval, portfolio optimization, machine learning prediction, and visualization.

The system architecture consists of the following major components:

- **Frontend Layer:** Developed using React.js for user interaction and dashboard visualization.
- **Backend Layer:** Developed using Flask for API handling, optimization logic, signal generation, and machine learning integration.
- **Database Layer:** PostgreSQL database used for user authentication and KYC management.

- **Financial Data Layer:** Yahoo Finance API integrated using the `yfinance` library.
- **Machine Learning Layer:** Ensemble prediction models implemented using Scikit-learn.

The frontend communicates with the backend through REST APIs using HTTP requests. The backend performs financial analysis, optimization, and machine learning predictions before sending processed results back to the frontend dashboard.

4.2 Authentication and KYC Implementation

The system implements secure authentication and KYC verification before allowing access to portfolio optimization features.

4.2.1 Signup Process

The signup module allows users to create accounts using:

- Name
- Email Address
- Password

Passwords are securely hashed before being stored in the PostgreSQL database. After successful registration, a JWT token is generated for authentication.

4.2.2 KYC Verification

After signup, users complete KYC verification by submitting:

- PAN Card
- Aadhaar Card

- Selfie Upload

The backend validates the JWT token and updates the KYC status in the database.

Only approved users are allowed to access portfolio optimization features.

4.2.3 Login Process

The login module performs:

- Credential verification
- Password validation
- KYC approval checking
- JWT token generation

Users with incomplete KYC verification are restricted from dashboard access.

4.3 Backend Implementation

The backend is implemented using Flask and is responsible for authentication, financial computations, portfolio optimization, machine learning prediction, and API handling.

4.3.1 Application Setup

The backend application is initialized in `app.py`. The backend integrates:

- Flask web framework
- Flask-CORS for frontend-backend communication
- SQLAlchemy ORM for database management
- JWT authentication manager

- Portfolio optimization modules
- Risk analysis modules
- Ensemble machine learning models

4.3.2 Implemented API Endpoints

The backend provides several REST API endpoints:

- `GET /` – Health check route
- `GET /stocks` – Returns supported NIFTY 50 stocks
- `POST /signup` – User registration
- `POST /login` – User authentication
- `POST /kyc/submit` – KYC verification
- `GET /profile` – User profile retrieval
- `POST /optimize` – Portfolio optimization
- `POST /ensemble/explain` – Ensemble prediction and explanation

All API responses are returned in JSON format.

4.4 Market Data Collection

Historical stock market data is collected using the Yahoo Finance API through the `yfinance` Python library.

The system:

- Downloads approximately two years of historical stock prices
- Supports NIFTY 50 stocks
- Removes invalid ticker symbols automatically

- Uses adjusted closing price for analysis

The collected data forms the base input for preprocessing, optimization, and prediction modules.

4.5 Data Preprocessing

Data preprocessing is implemented in `preprocess.py`.

The preprocessing module performs:

- Data cleaning
- Missing value handling
- Date alignment
- Daily return calculation

Daily returns are calculated using percentage price change between consecutive trading days.

The processed return matrix is then passed to risk analysis and optimization modules.

4.6 Risk Analysis Module

Risk analysis is implemented in `risk_metrics.py`.

The module calculates important financial metrics including:

- Mean returns
- Covariance matrix
- Portfolio volatility
- Sharpe ratio
- Value at Risk (VaR)

- Conditional Value at Risk (CVaR)
- Beta
- Maximum drawdown
- Risk contribution

These metrics help evaluate the stability and downside risk of the optimized portfolio.

4.7 Portfolio Optimization Module

Portfolio optimization is implemented in `portfolio_optimizer.py` using Modern Portfolio Theory.

The system receives:

- User selected stocks
- Investment amount
- Risk preference

The optimization module calculates optimal asset allocation weights using the Sequential Least Squares Programming (SLSQP) algorithm from SciPy.

4.7.1 Optimization Modes

The optimization strategy changes according to the selected risk profile:

- **Low Risk:** Minimize portfolio volatility
- **Medium Risk:** Maximize Sharpe Ratio
- **High Risk:** Maximize expected return

4.7.2 Optimization Constraints

The optimization process follows the constraints:

- Total portfolio weight must equal 1
- Asset weights must remain between 0 and 1
- Individual stock concentration limits are applied

The optimized portfolio allocation is then returned to the dashboard.

4.8 Trading Signal Generation

Trading signal generation is implemented in `trading_signals.py`.

The system generates BUY, HOLD, and SELL signals using technical indicators.

4.8.1 Technical Indicators Used

- Relative Strength Index (RSI)
- 20-Day Simple Moving Average (SMA20)
- 50-Day Simple Moving Average (SMA50)
- Momentum Indicator

4.8.2 Signal Rules

Signals are generated using:

- RSI < 35 indicates BUY signal
- RSI > 65 indicates SELL signal
- SMA20 $>$ SMA50 indicates bullish trend
- Positive momentum indicates upward trend

Multiple indicator outputs are combined to generate final stock recommendations.

Portfolio-level BUY, HOLD, or SELL recommendations are also generated using aggregate stock signals.

4.9 Ensemble Machine Learning Implementation

Phase 2 introduces ensemble machine learning for prediction and explainability.

4.9.1 Models Used

The ensemble pipeline combines predictions from:

- Ridge Regression
- Random Forest
- HistGradientBoosting

4.9.2 Prediction Workflow

The system extracts features such as:

- Recent returns
- Volatility
- Momentum
- Sector exposure
- Risk indicators

Predictions from all models are averaged using ensemble learning to generate the final prediction score.

The ensemble module also generates explainable prediction insights for users.

4.10 Frontend Implementation

The frontend dashboard is developed using React.js and Tailwind CSS.

4.10.1 Dashboard Features

The dashboard allows users to:

- Select NIFTY 50 stocks
- Choose investment amount
- Select risk level
- Run portfolio optimization
- View portfolio allocation
- Analyze risk metrics
- Compare portfolio against benchmark
- View trading signals
- Access ensemble prediction insights

4.10.2 Visualization Components

The frontend uses Recharts for visualization including:

- Portfolio allocation pie chart
- Sector allocation chart
- Risk contribution bar chart
- Portfolio versus benchmark graph
- Trading signal panels
- Performance metric cards

4.11 Database Implementation

The database configuration is managed using PostgreSQL and SQLAlchemy ORM.

The database stores:

- User account information
- Password hashes
- Authentication details
- KYC verification status

Database tables are initialized during backend setup.

4.12 Implementation Workflow

The complete workflow of the system is summarized below:

1. User signs up and completes KYC verification.
2. User logs into the dashboard.
3. User selects stocks, investment amount, and risk level.
4. Backend fetches historical stock data using Yahoo Finance.
5. Data preprocessing and return calculations are performed.
6. Risk metrics and covariance matrices are calculated.
7. Portfolio optimization is executed using SLSQP optimization.
8. Trading signals are generated using technical indicators.
9. Ensemble machine learning predictions are generated.
10. Portfolio metrics and visualizations are returned to the frontend.
11. Dashboard displays optimized portfolio recommendations and risk analysis.

4.13 System Interfaces

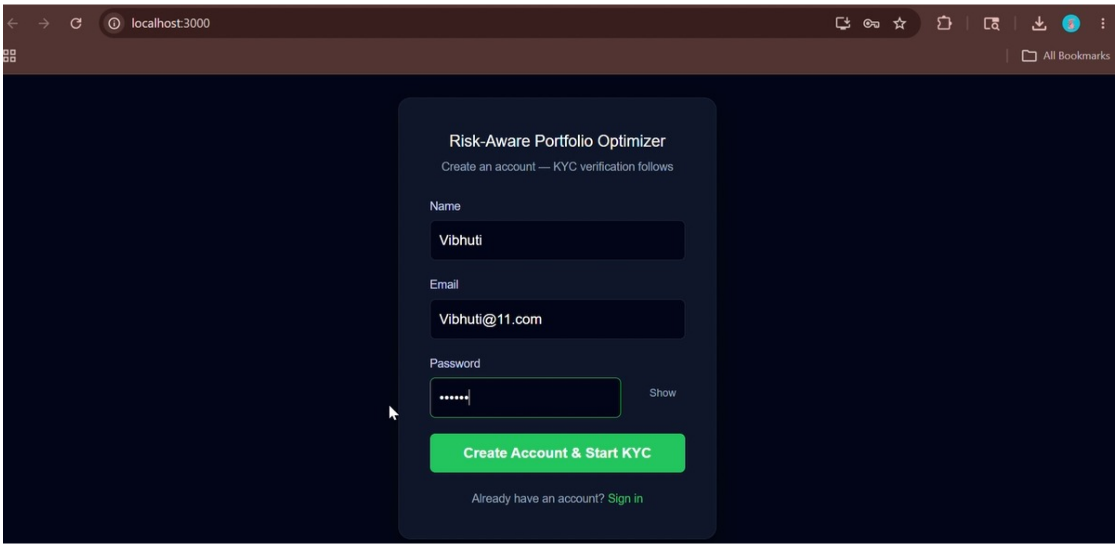

Figure 4.3: Sign up Page

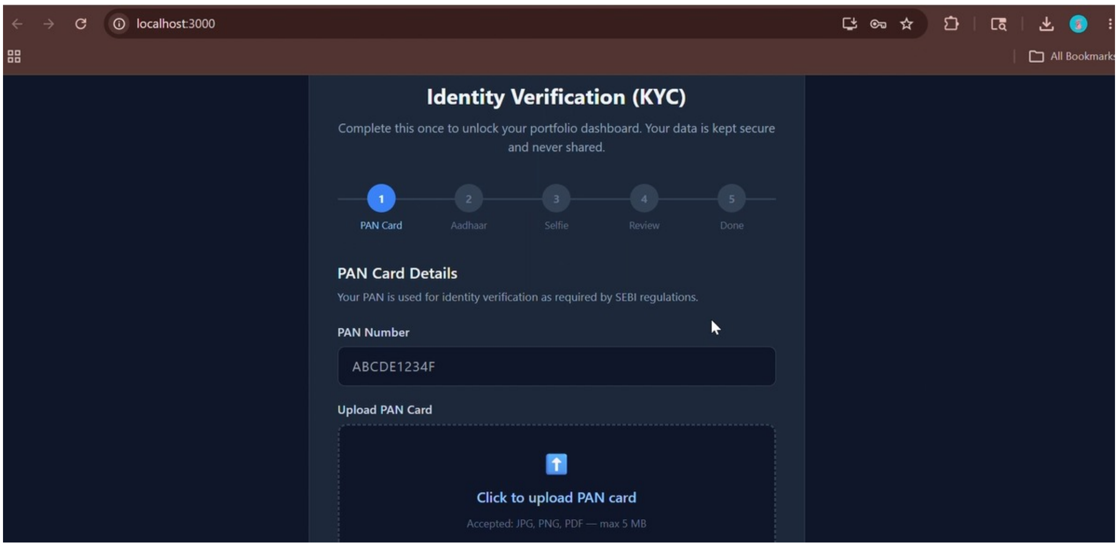

Figure 4.4: KYC Process Start

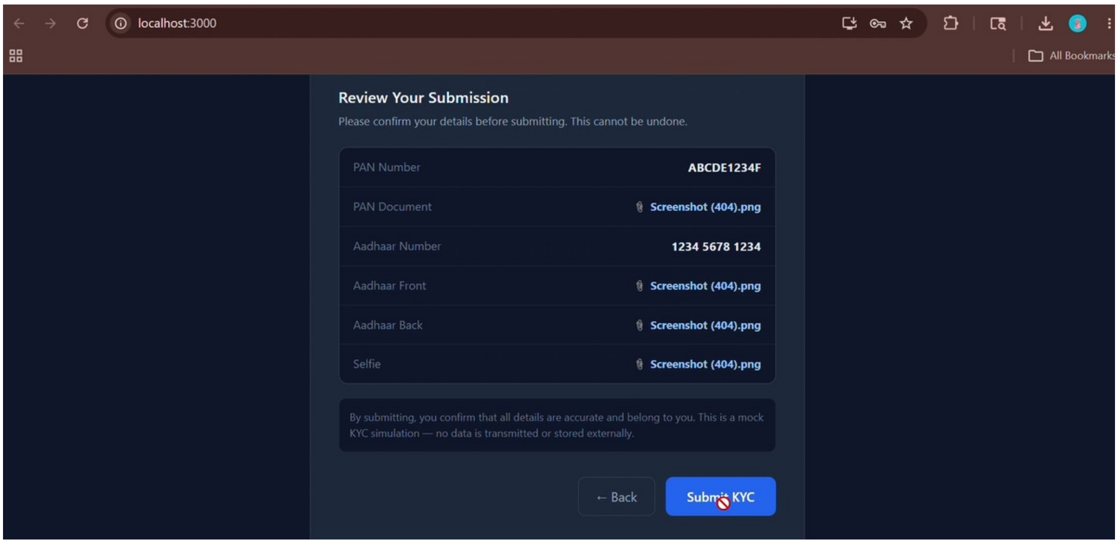

Figure 4.5: KYC Documents Review

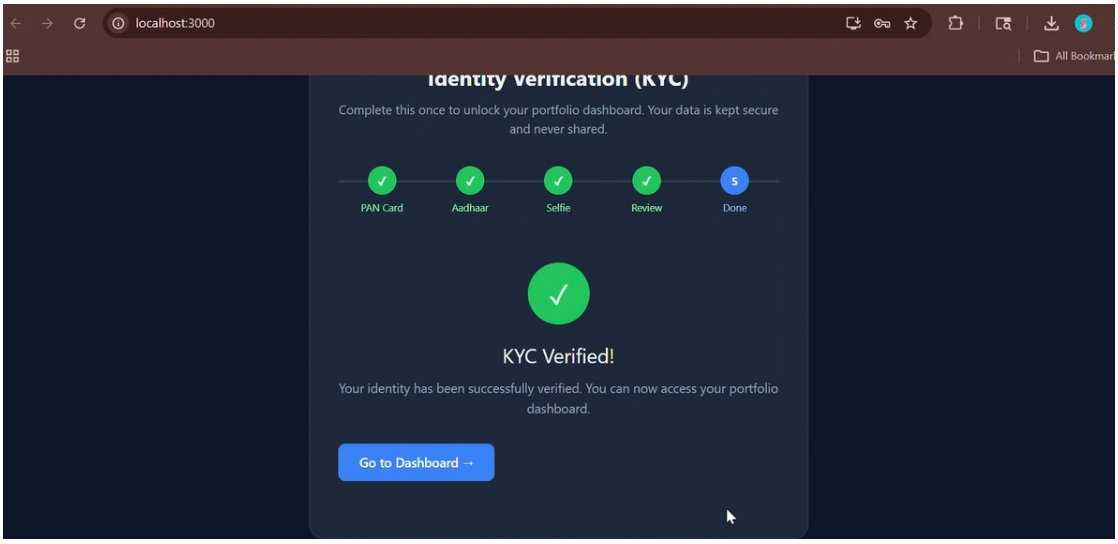

Figure 4.6: KYC Verified

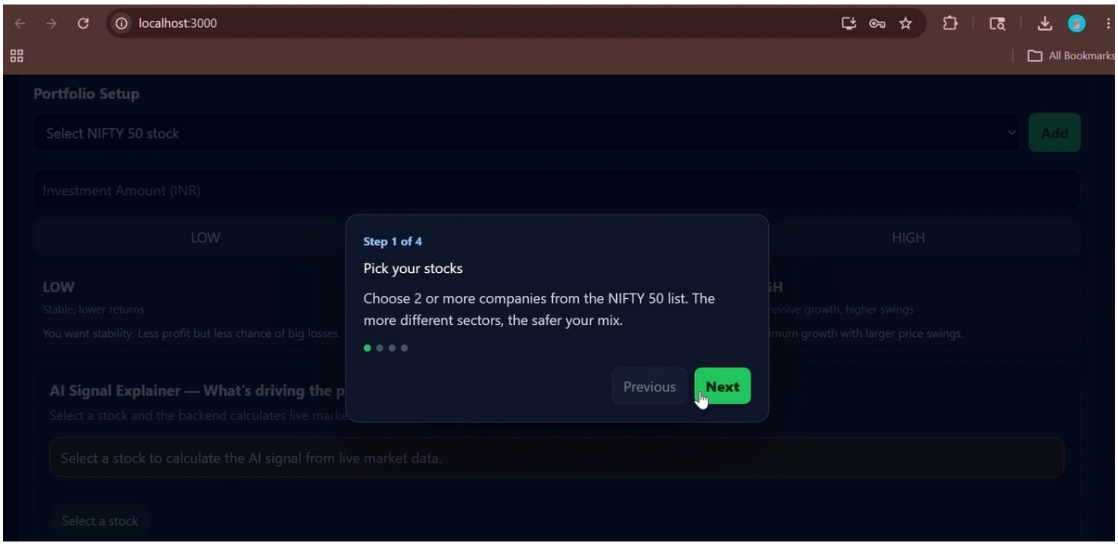

Figure 4.7: Step 1 Pick your stocks

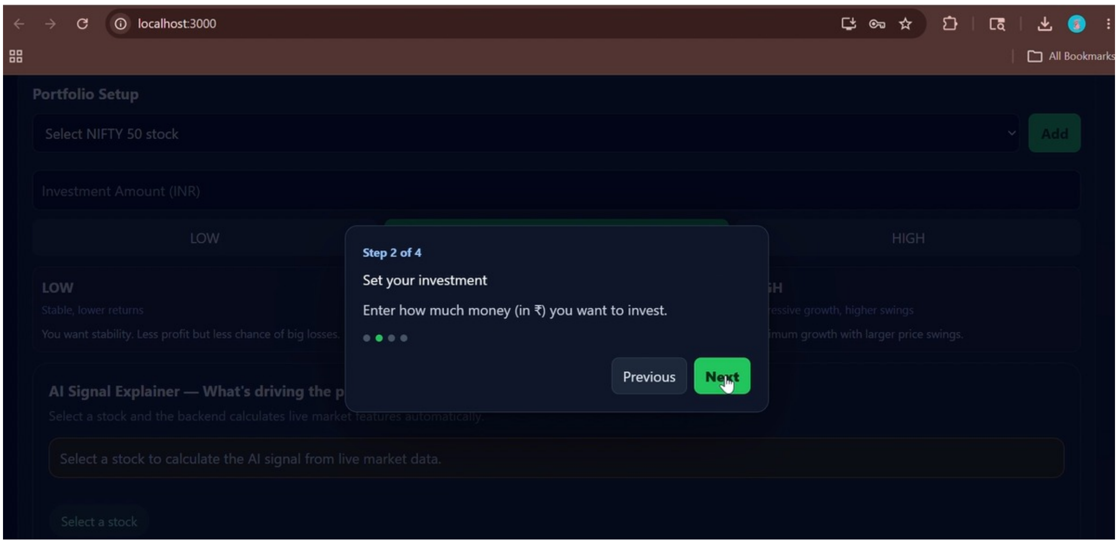

Figure 4.8: Step 2 Select your Investment

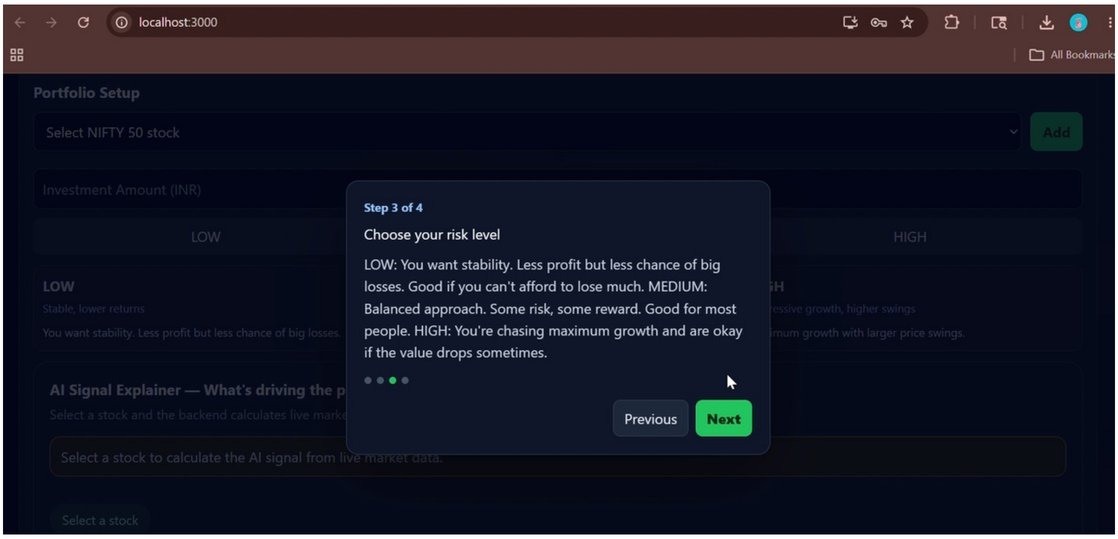

Figure 4.9: Step 3 Choose your risk level

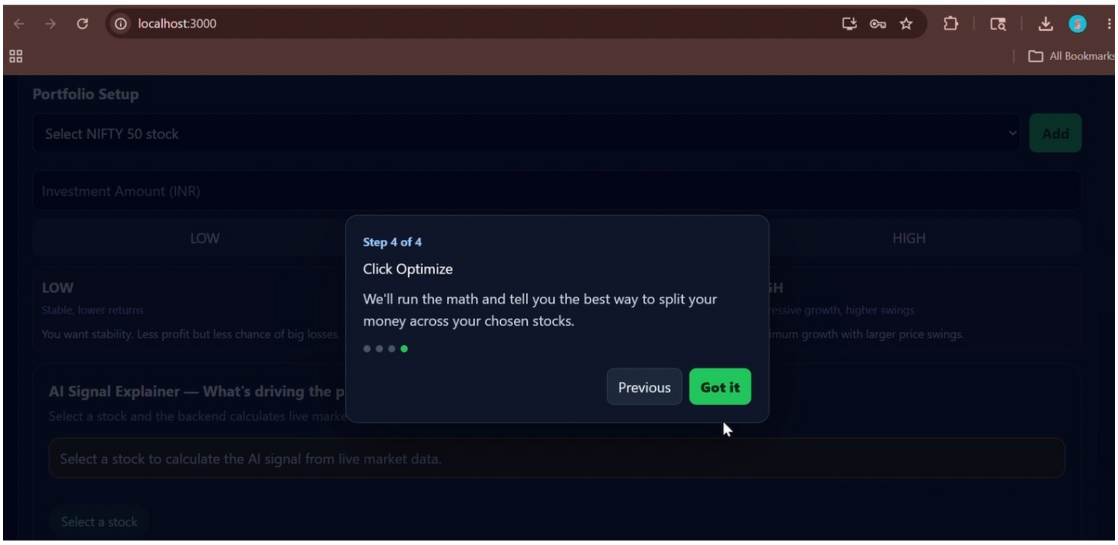

Figure 4.10: Click optimize

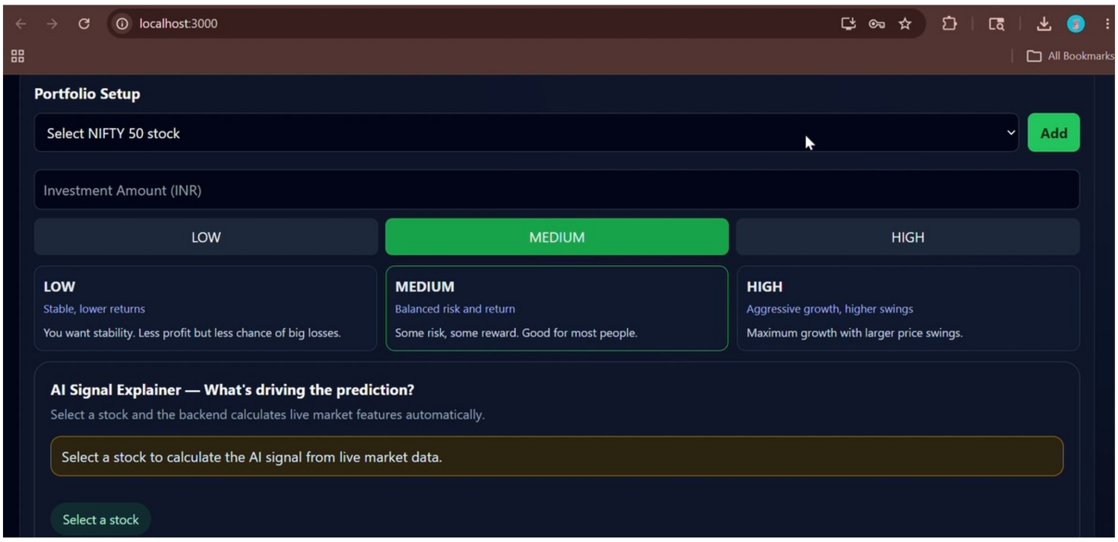

Figure 4.11: Dashboard Interface

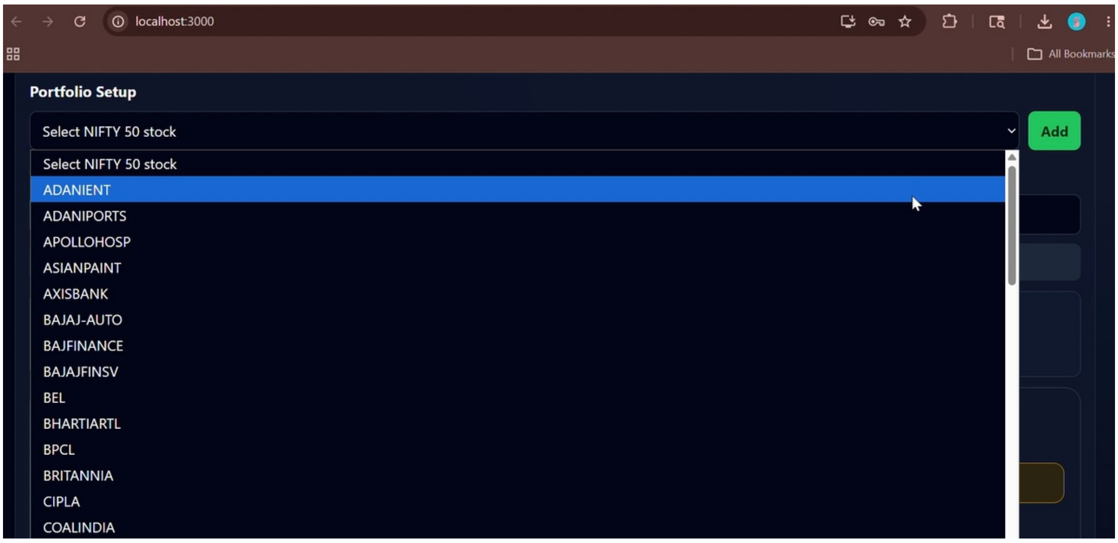

Figure 4.12: Select stocks

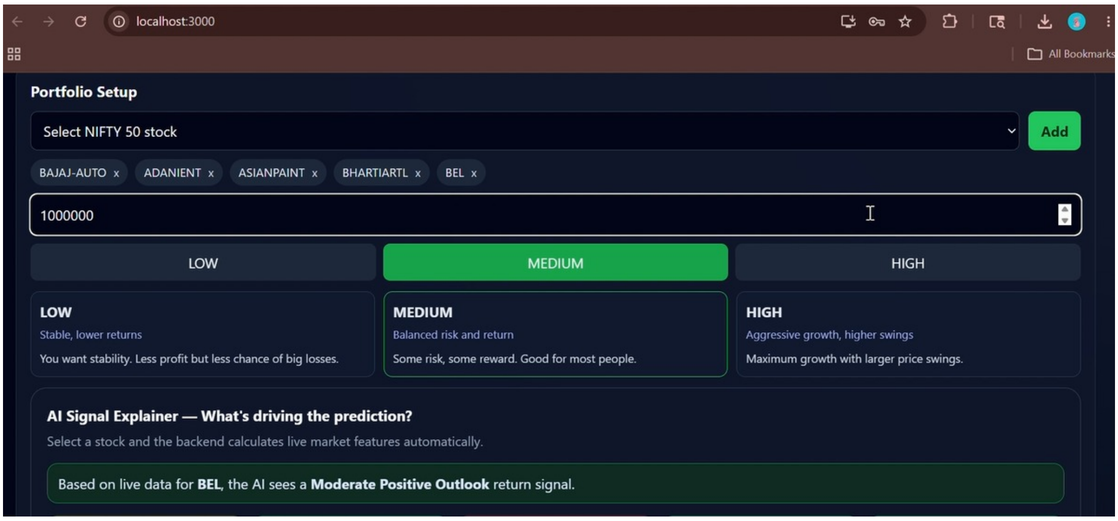

Figure 4.13: Select Risk Level

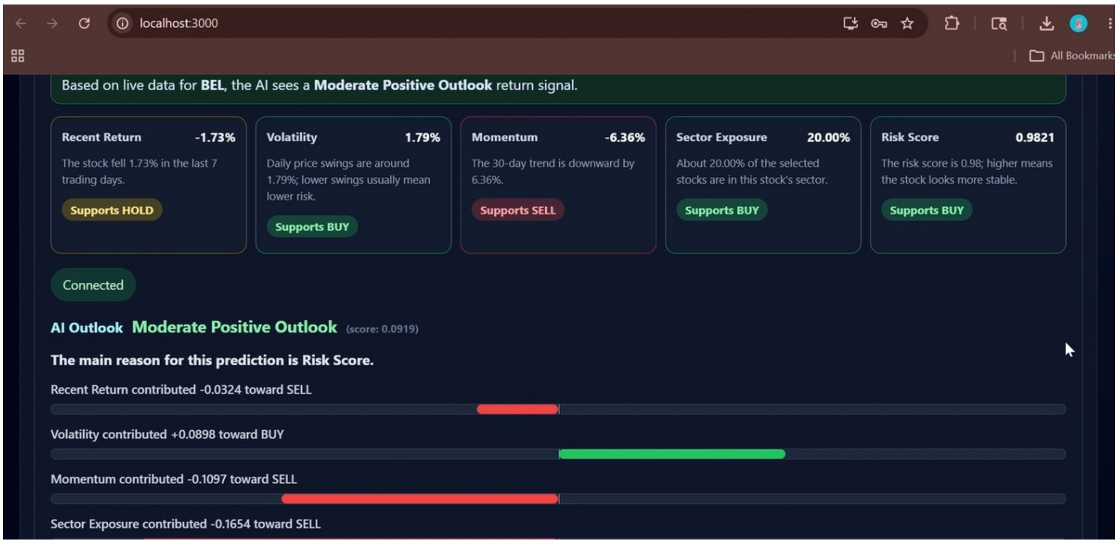

Figure 4.14: Indicators

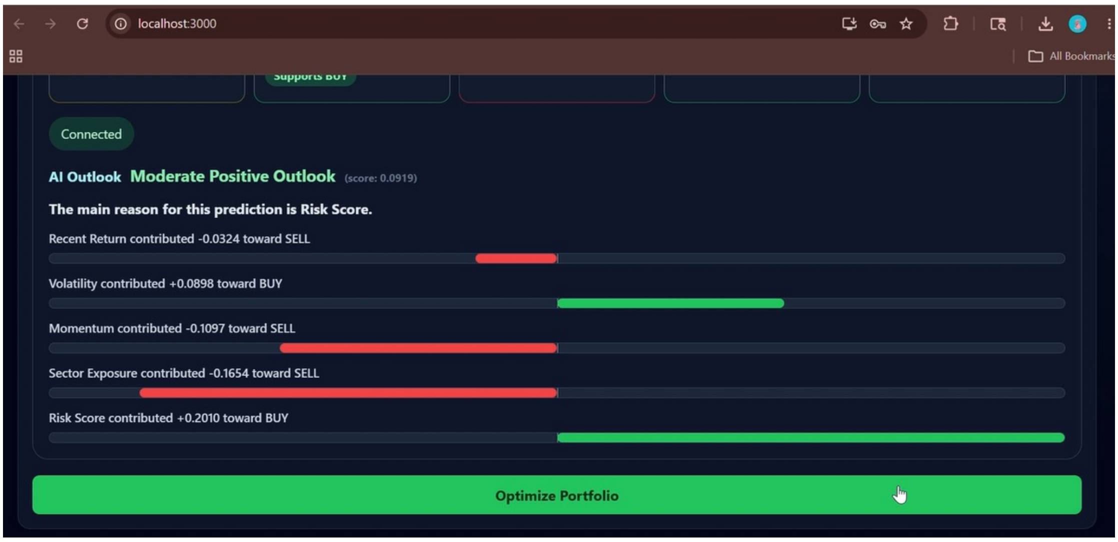

Figure 4.15: Explainable Insights

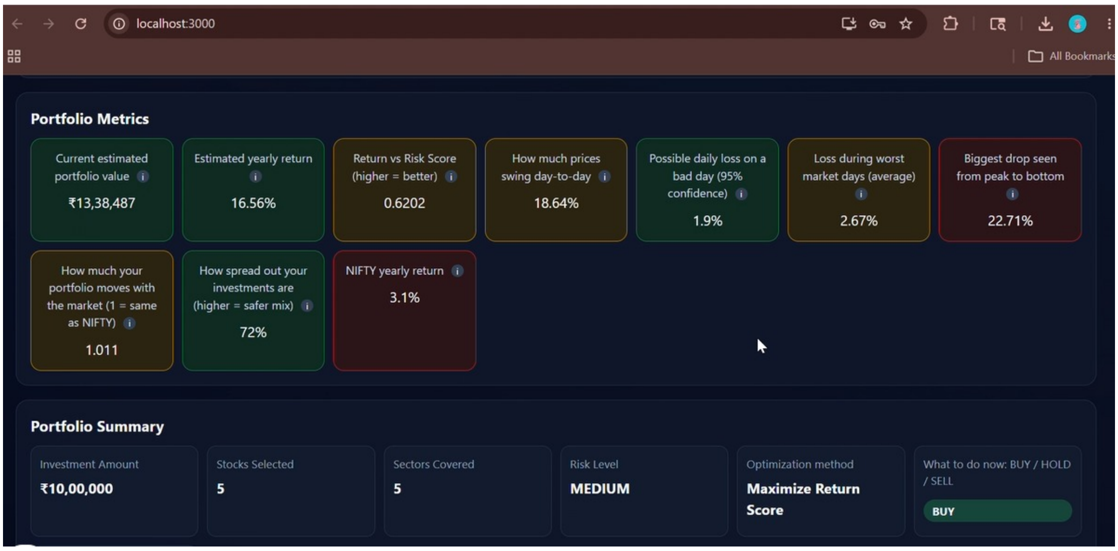

Figure 4.16: Portfolio Metrics Portfolio Summary

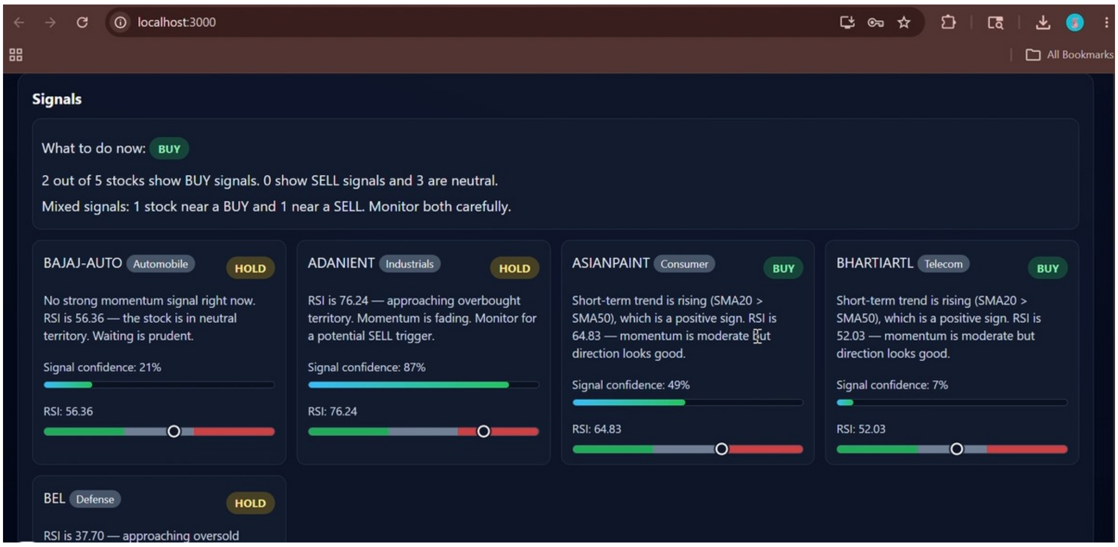

Figure 4.17: Signals Generated

Figure 4.18: Performance Chart

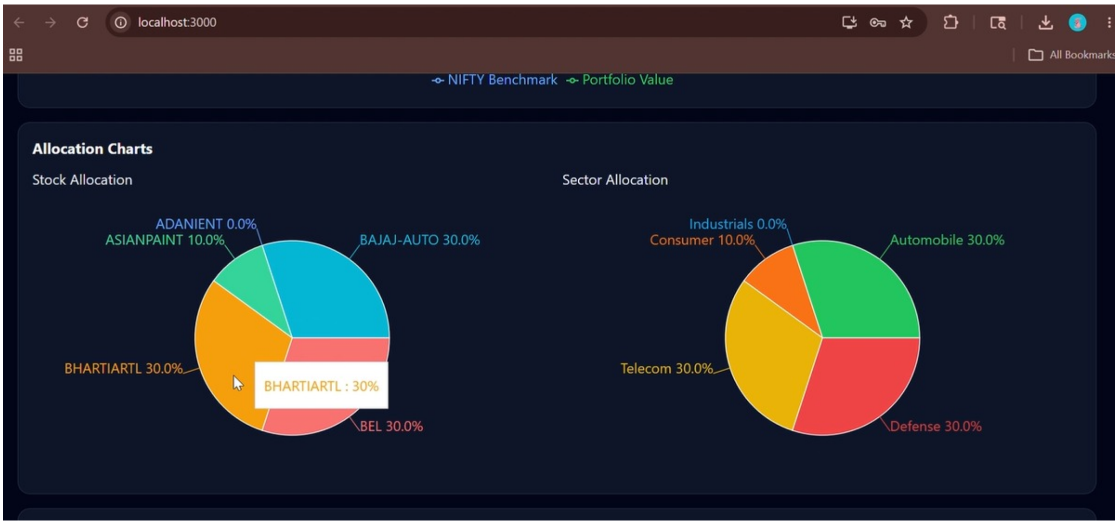

Figure 4.19: Allocation Chart

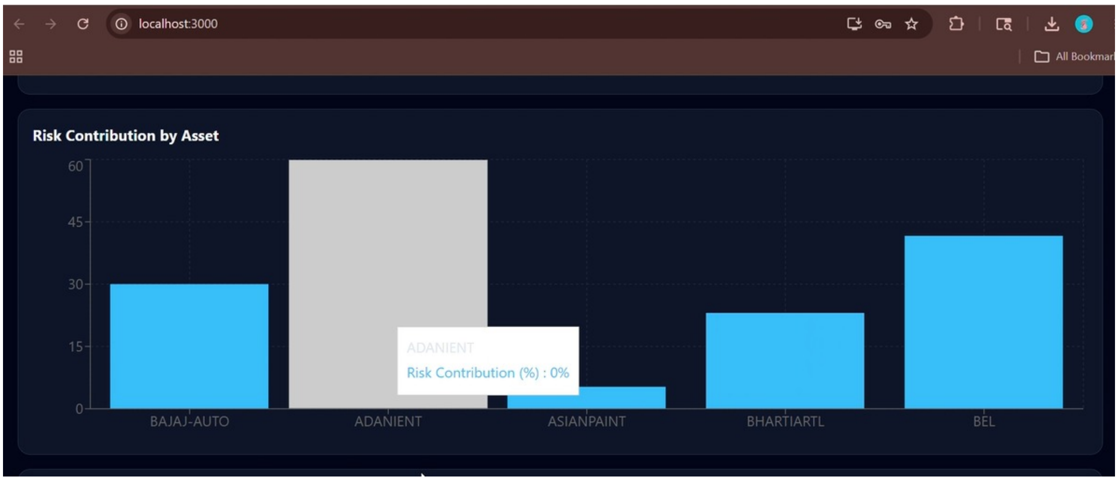

Figure 4.20: Risk Contribution by asset

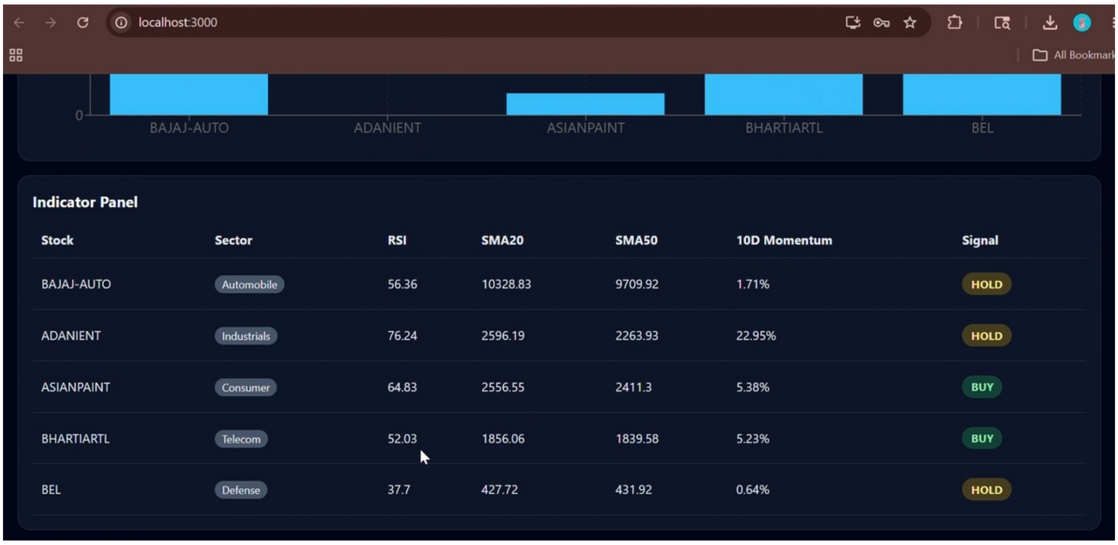

Figure 4.21: Indicator Panel

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The Risk-Aware Portfolio Optimization System for Stock Trading successfully demonstrates the integration of financial analytics, portfolio optimization, machine learning, and web technologies into a unified decision-support platform. The developed system provides users with an interactive environment for stock selection, risk analysis, portfolio optimization, and trading signal generation using real historical market data.

The implementation includes a Flask-based backend, React-based frontend, PostgreSQL database integration, and REST API communication between modules. Historical stock market data is collected using the Yahoo Finance API and processed using preprocessing and financial analysis techniques. The system calculates important portfolio risk metrics such as volatility, covariance, Sharpe ratio, Value at Risk (VaR), and Conditional Value at Risk (CVaR). Portfolio optimization is implemented using Modern Portfolio Theory and the Sequential Least Squares Programming (SLSQP) optimization algorithm to generate optimal asset allocation based on user-defined risk preferences.

The project also implements technical indicator-based BUY, SELL, and HOLD signal generation using RSI, SMA20, SMA50, and momentum indicators. In Phase 2, the system was further enhanced with KYC authentication workflow, improved dashboard visualization, benchmark comparison, and an ensemble machine learning pipeline using Ridge Regression, Random Forest, and HistGradientBoosting models. Ensemble predictions and explainable signal outputs improve the overall analytical capability of the system.

The developed prototype successfully demonstrates how portfolio optimization, risk management, technical analysis, and ensemble machine learning can be combined within a modern web-based platform to support intelligent and risk-aware investment decision-making.

5.2 Future Scope

Although the current system provides a functional and interactive portfolio optimization platform, several enhancements can be incorporated in future work to improve its performance, scalability, and practical applicability.

Future improvements may include:

- Integration of real-time stock market data streams for live portfolio monitoring
- Implementation of advanced Explainable AI techniques such as SHAP and LIME
- Integration of deep learning models such as LSTM for advanced stock trend prediction
- Deployment of the system on cloud infrastructure for scalable access
- Addition of portfolio history tracking and investment analytics
- Real-time alerts and notification systems for trading recommendations
- Support for cryptocurrency and forex portfolio optimization
- Integration of automated trading APIs for real-time execution
- Addition of advanced financial metrics such as Sortino Ratio and Maximum Drawdown

- Mobile-responsive and Progressive Web Application (PWA) support

These future enhancements would improve prediction accuracy, scalability, usability, and real-world applicability of the proposed system, making it more suitable for practical financial decision support and intelligent investment management.

Appendix A

Important Terms

This appendix describes important financial, optimization, and machine learning terms used in the project.

Portfolio Optimization

Portfolio optimization is the process of selecting the best combination of financial assets in order to maximize expected return while minimizing portfolio risk. In this project, portfolio optimization is implemented using Modern Portfolio Theory (MPT) and the SLSQP optimization algorithm.

Return

Return represents the percentage change in the price of a financial asset over a period of time. It indicates the profit or loss generated by an investment.

Volatility

Volatility measures the variability or fluctuation in asset prices over time. Higher volatility indicates higher investment risk.

Sharpe Ratio

The Sharpe Ratio is a risk-adjusted performance metric used to evaluate how much excess return a portfolio generates relative to the amount of risk taken.

Value at Risk (VaR)

Value at Risk estimates the maximum expected portfolio loss over a specified time period at a given confidence level.

Conditional Value at Risk (CVaR)

Conditional Value at Risk measures the expected loss during extreme downside market conditions beyond the VaR threshold.

Relative Strength Index (RSI)

RSI is a technical momentum indicator used to identify overbought and oversold market conditions.

Simple Moving Average (SMA)

A Simple Moving Average calculates the average price of a stock over a specified number of trading periods and helps identify market trends.

Momentum Indicator

Momentum measures the rate of change in stock price movement over time and helps identify upward or downward market trends.

Sequential Least Squares Programming (SLSQP)

SLSQP is a numerical optimization algorithm used to solve constrained optimization problems. It is used in this project to determine optimal portfolio allocation weights.

Diversification

Diversification is the process of spreading investments across different assets or sectors to reduce concentration risk in a portfolio.

Ensemble Machine Learning

Ensemble learning combines predictions from multiple machine learning models to improve prediction stability and reduce model bias.

Random Forest

Random Forest is an ensemble machine learning algorithm based on multiple decision trees used for prediction and classification tasks.

Ridge Regression

Ridge Regression is a linear regression technique that applies regularization to reduce overfitting and improve prediction stability.

HistGradientBoosting

HistGradientBoosting is a gradient boosting machine learning algorithm optimized for faster training and improved prediction performance.

JWT Authentication

JSON Web Token (JWT) authentication is a secure token-based authentication mechanism used for verifying user identity and protecting backend APIs.

KYC Verification

Know Your Customer (KYC) verification is the identity validation process implemented in the system using PAN card, Aadhaar card, and selfie verification before allowing dashboard access.

Appendix B

Mathematical Formulations

This appendix summarizes the important mathematical models and financial equations used in the proposed portfolio optimization system.

Daily Return

The daily return of a stock is calculated as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{B.1}$$

where:

- P_t is the stock price at time t
- P_{t-1} is the stock price at time $t-1$

Expected Return

The expected return of an asset is calculated as the average historical return:

$$\mu = \frac{1}{N} \sum_{t=1}^{N} r_t \tag{B.2}$$

where:

- N is the total number of observations
- r_t is the return at time t

Covariance Matrix

The covariance matrix measures the relationship between returns of different assets:

$$\Sigma_{ij} = \frac{1}{N-1} \sum_{t=1}^{N} (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j) \tag{B.3}$$

where:

- $r_{i,t}$ is the return of asset i
- $r_{j,t}$ is the return of asset j
- $\bar{r}_i$ and $\bar{r}_j$ are mean returns

Portfolio Return

The expected return of the portfolio is calculated as:

$$R_p = \sum_{i=1}^{n} w_i \mu_i \tag{B.4}$$

where:

- w_i represents the weight of asset i
- μ_i represents the expected return of asset i

Portfolio Volatility

Portfolio risk is calculated using:

$$\sigma_p = \sqrt{w^T \Sigma w} \tag{B.5}$$

where:

- w is the portfolio weight vector
- Σ is the covariance matrix

Sharpe Ratio
The Sharpe Ratio measures risk-adjusted return:

$$Sharpe = \frac{R_p - R_f}{\sigma_p} \tag{B.6}$$

where:

- R_p is portfolio return
- R_f is the risk-free rate
- σ_p is portfolio volatility

Value at Risk (VaR)
Value at Risk at 95% confidence level is estimated as:

$$VaR_{95} = -Q_{5\%}(R) \tag{B.7}$$

where $Q_{5\%}(R)$ represents the 5th percentile of portfolio returns.

Conditional Value at Risk (CVaR)
Conditional Value at Risk is calculated as:

$$CVaR_{95} = -E[R \mid R \leq Q_{5\%}(R)] \tag{B.8}$$

Relative Strength Index (RSI)
The RSI indicator is calculated using:

$$RS = \frac{\text{Average Gain}}{\text{Average Loss}} \tag{B.9}$$

$$RSI = 100 - \frac{100}{1 + RS} \tag{B.10}$$

Simple Moving Average (SMA)
The Simple Moving Average is calculated as:

$$SMA_n = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{B.11}$$

where P_i represents stock prices over n periods.

Momentum Indicator

Momentum is calculated using price difference over a fixed interval:

$$Momentum = P_t - P_{t-n} \tag{B.12}$$

where:

- P_t is the current price
- P_{t-n} is the historical price

Diversification Score

Sector diversification is measured as:

$$Diversification = \left(1 - \sum_{j=1}^{k} s_j^2\right) \times 100 \tag{B.13}$$

where:

- s_j represents sector allocation weight
- k represents total sectors

Ensemble Prediction

The final ensemble prediction is calculated as:

$$Prediction = \frac{P_{RF} + P_{RR} + P_{HGB}}{3} \tag{B.14}$$

where:

- P_{RF} is Random Forest prediction
- P_{RR} is Ridge Regression prediction
- P_{HGB} is HistGradientBoosting prediction

References

[1] H. Markowitz, “Portfolio selection,” *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.

[2] W. F. Sharpe, “Capital asset prices: A theory of market equilibrium under conditions of risk,” *The Journal of Finance*, vol. 19, no. 3, pp. 425–442, 1964.

[3] E. F. Fama, “Efficient capital markets: A review of theory and empirical work,” *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.

[4] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. Holden-Day, 1970.

[5] G. Zhang, B. E. Patuwo, and M. Y. Hu, “Forecasting with artificial neural networks,” *International Journal of Forecasting*, 1998.

[6] K. Kim, “Financial time series forecasting using support vector machines,” *Neurocomputing*, 2003.

[7] L. Breiman, “Random forests,” *Machine Learning*, 2001.

[8] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, 1997.

[9] “Study of existing platforms in stock trading and portfolio optimization,” 2024.

[10] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *Journal of Econometrics*, 1986.

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Date:

Name of Candidate